

PROJECT II—ASIAN CALL AND PUT OPTIONS FOR CRYPTOCURRENCIES

1 Objective

Price Asian call and put options (fixed strikes) with Bitcoin / ETH as the underlying asset.

2 Data source

- Bitcoin / ETH price:
<https://finance.yahoo.com/quote/BTC-USD/history?p=BTC-USD>;
<https://finance.yahoo.com/quote/ETH-USD/history?p=ETH-USD>
- Options data: <https://www.deribit.com/options/BTC/BTC-30MAY25>

3 Questions

1. Download underlying asset's price data, analyze the data, estimate log return mean, volatility, and other parameter values. You may consider two risk-free rates: $r = 1\%$ and $r = 5\%$ in your project.
2. Use standard binomial tree method (BTM) to price Asian call and put options in European style, including Asian arithmetic and Asian geometric. Report the numerical results for different number of periods, strikes, Bitcoin/ETH prices, and time to maturities.
3. Use modified BTM (Hull & White's method or Forward shooting grid method) to price Asian call and put options in European style. Report the numerical results for different strikes, Bitcoin/ETH prices, and time to maturities.
4. Compare the convergence and computing efficiency of standard BTM and modified BTM.
5. Risk management analysis: compute the Greeks of Asian options for both Bitcoin and ETH and compare the Greeks for Bitcoin and for ETH.
6. Note that there are only vanilla options (European options) on Bitcoin/ETH traded in the market. To persuade a platform (i.e., Deribit) to launch Asian options and to persuade investors to trade Asian options, please provide analysis on the pros and cons of Asian options for Bitcoin/ETH. Justify your analysis with numerical results. For instance, you may use your pricing algorithm to provide comparison analysis between Asian options and Vanilla options (for simplicity, you may use Black-Scholes formula to price vanilla options numerically).
7. Bitcoin / ETH prices have many jumps, will hedging with Asian options help to mitigate such risk?

4 Requirements on project report

1. Introduction: a summary of research objective, methodology, and findings
2. Methodology and data: the pricing method, data description, and estimation method used in your project
3. Project findings: data analysis, pricing results and analysis, and discussions.
4. Conclusion.